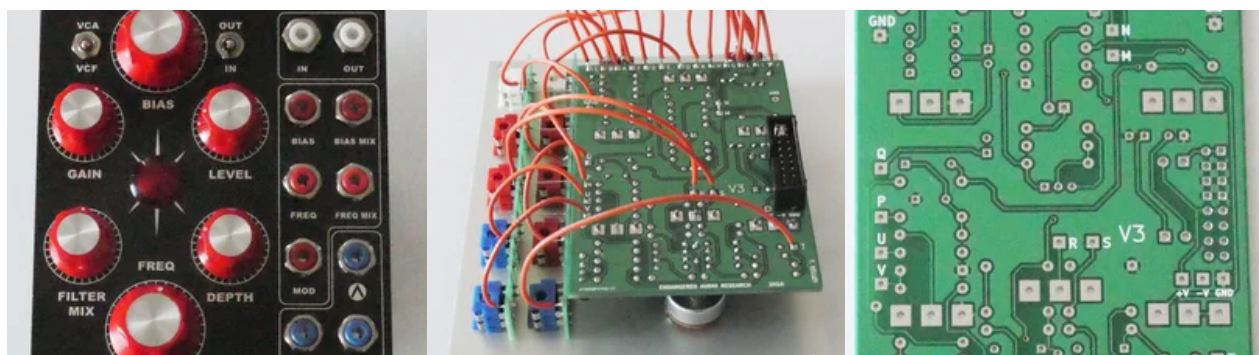


Gristleizer Synthesizer Module Build Guide

By [endangeredaudioresearch](#) in [CircuitsElectronics](#)

Introduction: Gristleizer Synthesizer Module Build Guide



This instructable will guide you through construction of the Endangered Audio Research Gristleizer from a kit or PCB. Steps that are kit or PCB only will be marked (Kit) or (PCB) respectively.

Although this version is intended for Eurorack, you can also install it in 5U, MOTM, Serge, or other systems as it runs on +/- 12-15V.

The Gristleizer is a legendary VCA/VCF effect used extensively by industrial music pioneers Throbbing Gristle. Developed by a 15 year-old Roy Gwinn and published in a DIY electronics magazine in 1975, the original design found its way to Throbbing Gristle, who brought the Bias trimpot to the front panel and dubbed it The Gristleizer.

Decades later, Endangered Audio Research released the first production Gristleizer tabletops, which faithfully recreated the circuit while solving many of the problems inherent in the original design.

Now, after 8 years of continuous refinement, including a major overhaul of the filter circuit in version 3, expanded controls, and more CV options, Endangered Audio Research presents the Eurorack version of The Gristleizer. This is an extremely limited run, after which The Gristleizer name will be retired from the Endangered Audio Research product line permanently.

Features:

Knobs:

BIAS - In VCA mode, the Bias effects the intensity and shape of the modulation, amount of distortion, and overall level of the modulated signal. In VCF mode, the Bias sweeps the center frequency of the

improved bandpass filter.

GAIN - Controls the amount of signal drive in the first gain stage.

FILTER MIX - Controls the amount of dry signal added to the filtered signal.

FREQ - Controls the frequency of the internal LFO.

DEPTH - Controls the amount of LFO modulation applied to the signal.

LEVEL - Controls the output level of the module.

Switches:

MODE SWITCH - Selects VCA or VCF mode.

IN/OUT SWITCH - True bypass switch.

Light:

LED - Indicates the rate of the internal LFO. A specially selected Fresnel lens and LED provides a convincing incandescent look.

Jacks:

IN - Signal to be Gristleized.

OUT - Gristleized signal (when effect is switched in).

BIAS IN - Allows for external control of the Bias voltage. Can be modulated by any signal of sufficient strength. This is a switching jack - when patched, it disconnects the BIAS knob on the front panel.

BIAS MIX IN - Same as the BIAS IN, but does not disconnect the front panel BIAS control. Instead, the external input is mixed with and offset by the BIAS knob.

FREQ IN - Allows for external control of the LFO frequency. This is a switching jack - when patched, it disconnects the FREQ knob on the front panel.

FREQ MIX IN - Same as the FREQ IN, but does not disconnect the front panel FREQ control. Instead, the external input is mixed with and offset by the FREQ knob.

MOD IN - This switching jack is normalled to the Triangle LFO, meaning you do not need to patch the module for it to function. Inserting a patch cable will break this connection and allow you to modulate The Gristleizer with any modulation source, including LFOs, envelopes, sequencers, sample & hold, and audio.

TRIANGLE OUT - Triangle LFO SAWTOOTH OUT - Saw LFO SQUARE OUT - Square LFO - not a true square wave, but optimized with sloped rising and falling edges to prevent the clicking that plagues the original design.

Step 1: (Kit) Unpack and Sort Components

Unpack the kit and sort the components. For small components like resistors and capacitors, group similar parts together and put them on a piece of paper. Mark the value of each group of components on the paper next to them. If you cannot read resistor codes (pictured above) or do not have a multimeter, you will not be able to complete your build.

Step 2: (PCB) Purchase Components (Bill of Materials, Component Reference, & Schematic)

Control Panel Drill Template

Resistors (Metal Film):

NOTE: R3 is 22k, *not* 47k (this was changed after boards were printed)

Value - Quantity

- 22 - 1
- 1k - 5
- 2.2k - 1
- 4.7k - 1
- 10k - 6
- 18k - 2
- 22k - 1
- 27k - 2
- 33k - 2
- 47k - 3
- 56k - 2
- 82k - 1
- 100k - 4
- 180k - 1
- 470k - 1
- 1m - 2

Capacitors:

Value - Quantity - Package Code

- .002uF (2nF) - 1 - 222
- .01uF (10nF) - 2 - 103
- .1uF (100nF) - 2 - 104
- .47uF (470nF) - 1 - 474

Electrolytic Capacitors:

Value - Quantity

- 1uF - 1
- 10uF -2
- 100uF - 2

Diodes:

Type - Quantity

- 1N4148 - 2
- 1N4738 - 1
- 1N5817 - 2
- LED - 1

Transistors:

Type - Quantity

- 2N3906 - 2
- BF256 - 1

ICs:

Type - Quantity

- RC4558 - 3

Potentiometers:

Value - Type - Quantity

- 10k - Logarithmic (A) - 1
- 100k - Linear (B) - 5

Switches:

Type - Quantity

- DPDT - 2

Jacks:

Type - Quantity

- 3.5mm Mono Switching - 10

Wire:

Gauge - Length

- 24awg - 6 feet (2 meters)

Knobs:

Type - Quantity

- Small - 4
- Large - 2

LED Lens:

Type - Quantity

- Fresnel Lens - 1

Power Socket:

Type - Quantity

- Modular Keyed - 1

Step 3: (PCB) Parts Sourcing Suggestions

We source most parts from www.mouser.com. For international builders, we suggest using www.farnell.com.

All **resistors** are 1/4 watt metal film 1% tolerance.

Electrolytic & film capacitors must be rated at 25 volts or more. Be sure to check the lead spacing dimension of the part and make sure it will fit the component footprint on the PCB.

For **potentiometers**, we use 16mm Alpha long pin right angle PCB mount pots. These are available from [Small Bear](#). You can, of course, use your own potentiometers, especially if you do not plan to mount the PCB parallel to your panel.

The **3.5mm jack** we use is custom made, but you can use any 3.5mm jack (be aware: you may need to change the drill template to match your part). We use switching jacks, but they are not necessary unless you would like to normal the connection between the triangle wave and Mod input. A suitable jack can be found here on [MCM Electronics](#).

For the **LED**, you can use almost any standard one with or without a lens.

The **Power socket** can be any brand that you like, so long as it has the standard .1" (2.54mm) pin spacing. You may also want to directly wire to the V+, V-, and GND connections, especially if you are not building a Eurorack Gristleizer. We use a standard 16 pin socket that is available from [Mouser](#).

For the **power supply**, you can run this board at +/- 12V or +/-15V.

Step 4: Cut, Strip, and Tin Wires

Cut and strip the wires according to the length and the quantities listed below, and then set them aside. Strip the end of each wire approximately 3/16ths of an inch (5mm). We suggest you tin them by melting a little solder onto the tip of each wire — this prevents fraying as they are inserted into the switches and PCBs.

- 2in (5cm): 13 wires
- 3in (7.5cm): 4 wires
- 4in (10cm): 3 wires

Step 5: Populate and Solder Resistors

Populate and then solder all resistors. Before soldering, double and then triple check that you are putting the correct value into the correct space. Once they have been soldered, cut the resistors' leads and set them aside to be used as jumpers. We recommend you use a multimeter to check resistor values - if you do not have a multimeter, use the color codes above.

Step 6: Connect the M and N Pads With a Jumper

Use one of the resistor leads you saved as a jumper from pads M to N on the back of the PCB. The loop of the jumper should stick out on the back of the PCB — meaning when you solder it, you should solder it on the component side of the board. You should do this because the jumper is a bare wire and you want to minimize the chance of it shorting to another component. Clip the leads of the jumper on the component side of the board and discard them.

Step 7: Populate and Solder Diodes

Populate all diodes, making sure to observe correct orientation. The stripe on the diode should match the stripe on the board. Not orienting diodes correctly is the #1 mistake DIY builders make. Once you have populated the diodes and checked their orientation, solder and then cut their leads. The diodes with thick leads may be difficult to solder because their leads act like heat sinks. To make them easier to solder, after inserting them into the board, bend their leads outward a little to keep them in place, and then cut most of the excess lead away before soldering. After you have soldered them, cut the remaining excess.

Step 8: Populate and Solder ICs

Populate the three RC4558 op-amps. Double and then triple check your orientation. The Texas Instruments RC4558 has a dot next to pin 1. Pin 1 is on the left side of the keyhole on the PCB silkscreen. This is another very common mistake that even experienced DIYers make, so slow down, focus, and put them in the correct way. Once you are certain that you have them oriented correctly, solder them to the board. ICs can be very heat sensitive, so when in doubt, wait 5-10 seconds between soldering each pin, or rotate between each IC to give the package time to cool off.

Step 9: Populate and Solder Transistors

Populate all transistors, again, making sure to observe correct orientation. Incorrectly orienting transistors is another popular DIY mistake. Look closely at each silkscreened outline: the long flat face of the outline should match the flat face of each transistor. Transistors do not need to be mounted flush with the PCB - push them in as far as they'll go and then while the board is still component side up, bend the transistors' leads outward so that they will stay in place when you flip the board over to solder them. After double and then triple checking your orientation, solder and then cut the transistors' leads.

Step 10: Populate and Solder Non-polarized Capacitors

Populate all the small, non-polarized capacitors. Double and triple check the values by reading the package code and checking it against the Bill of Materials. Solder and cut their leads.

Step 11: Populate and Solder Electrolytic Capacitors

Populate all the polarized electrolytic (barrel) capacitors. The stripe with the minus (-) symbol is the negative lead. Negative electrolytic leads are also often shorter than positive leads. Double and then triple check each capacitor's orientation. Once you are sure each capacitor is oriented correctly, solder and cut their leads.

Step 12: Populate and Solder Power Connector

Place the power connector on the back (solder side) of the PCB with its locating notch facing towards the center of the PCB as seen in the photo. You may find it helpful to use a clothespin to hold the connector in place as you solder. You should only need to solder one or two pins to keep the power connector in place, after which you can remove the clothespin and finish soldering. Be careful: applying heat too long to any one pin may melt its plastic casing. If you think you are soldering too slowly, let the component cool off between each solder pin for 5-10 seconds. Note: the power connector for this module is not "Red stripe down." Instead, the negative voltage enters at the top of the power connector. If you orient the power connector correctly, this won't be a problem because the keyed slot will prevent backwards connection. The power pads you see at the bottom of the connector are alternate pads for those who want to install this board into 5U or other non-Eurorack systems. The power inputs are polarity protected by diodes.

Step 13: Snap Off Locking Tabs From Pots

Break off the locking tabs on the potentiometers using wire cutters or pliers. The metal is soft and should give easily and break cleanly. Hold the body of the potentiometer with one hand (as shown) rather than holding the potentiometer by its pins.

Step 14: Note Potentiometer Orientation

Note the orientation of the potentiometers in the photo before proceeding. If the board is component side up, the “Endangered Audio Research” label should be at the top of the board. Going from top to bottom, the first potentiometer is casing first; then pins, while all the rest on the board are pins first; then casing.

Step 15: Populate and Solder Pots

Populate the PCB-mounted potentiometers on the component side. Double and then triple check each pot's value and orientation. Turn the board over solder one leg of each pot. After you apply the solder, check that each pot is flush with the component side of the board. If not, reheat the solder and with your free hand, reposition the pot. When you are confident that all pots are sitting flush on the PCB, solder the remaining pins.

Step 16: Populate and Solder the LED

Insert the LED (D4), making sure to double and then triple check its polarity. The flat edge of the LED lines up with the flat edge on the PCB silkscreen. Like the electrolytic capacitors, the shorter lead of the LED is negative. The LED should stand straight about a centimeter off the board. If you want, you can cut a straw a desired length and feed one of the LED's leads through it so that it stands straight at the correct height. The straw will also prevent accidental shorting from a rough hand during final assembly. Just make sure it is not too tall that it won't fit properly into the panel lens. Once you are certain of the LED's orientation and height, solder and cut the leads.

Step 17: (PCB) Design and Drill Your Panel

Attached is a .zip file that includes a .pdf and .ai version of the Gristleizer drill template.

There's no right way to design your panel; just make sure that the drill centers are accurate and that the labels you make for controls will not be covered by whatever type of knob you choose.

Following the drill template and using PCB mounted pots will result in a very skiff-friendly module.

If you are crafty, you can of course create a smaller, tighter layout with the PCB mounted at a right angle to the panel, but the result will not likely fit in shallow synth cases.

Step 18: Mount Lens to Panel

If you purchased a kit, carefully peel the protective plastic from the control panel. Next, mount the indicator light lens. The tolerance on the lens is very tight and you may need to push very hard to get it to pop through. If you cannot seat the lens with your hands alone, lay a towel on a clean table and push the panel down onto the lens with both hands, taking care not to bend the panel. Be aware of what the panel's face is touching to avoid scratches.

Step 19: Mount Board to Panel

Insert the potentiometer shafts through the front panel holes. You may need to bend and straighten the pots a little. The important thing here is to not jam them into place. You also want to make sure that the LED is seating correctly in the lens. Once all of the pots are seated and the LED is standing straight under the lens, fasten the potentiometers with their washer and nut. Be careful not to over-torque them: they only need to be hand-tight. Once you've finished that, set the module aside for a moment.

Step 20: Gather Switches and Wire

Gather the two switches along with 10 of the 2in (5cm) wires that you made in step 3 and one jumper you collected from the resistors in step 4. A vice or some way to hold the switch stable while you solder is highly recommended, but not required.

Step 21: Prepare Switches for Soldering

Take all the hardware off the switch except for the bottom nut and set that aside in a place where you won't easily lose them.

Step 22: Solder Wires to the Mode Switch

For the Mode switch (marked VCA/VCF on the panel), solder 6 of the 2in (5cm) wires, starting at the bottom of the switch and working your way up (see illustration below). Take special care not to overheat the switch - using a broad-tipped soldering iron and pre-tinned wires will go a long way to quickly and cleanly soldering the switch without conducting too much heat into its plastic housing. When in doubt, wait 5-10 seconds between each wire to allow the switch to cool down.

Step 23: Solder Wires to the Bypass Switch

For the Bypass switch, first solder the jumper between the bottom two pins, then solder the 4 remaining 2in (5cm) wires to the top four pins (see illustration below). Cut the excess lead from the jumper.

Step 24: Mount Switches to the Panel

Replace the star washer on the switch bushing and make sure the teeth of the washer are facing towards the panel so that they will catch and dig into the panel. Use the nut on the bottom of the switches' bushing to position them so that just enough threading (about 4-6 turns) is visible on the front panel. Hold the switch in one hand while tightening the front panel nut with the other. If you use a tool to tighten these switches, make sure you place a piece of paper between the tool and the panel so that you don't make any scratches.

Step 25: Solder Switches to the PCB

Now that the switches have been mounted, connect each wire to the board as they are labelled in the illustrations. Pay special attention to this so that the C pin of the Bypass switch goes to point C on the top of the board, and so on. Double and then triple check your wiring before moving on to the next step.

Step 26: (Kit) Prepare Jack Boards

If you have a vice, put the module in the vice with a cloth or paper towel around the panel and then into the vice to prevent scratching. Tighten just enough to hold it steady. If you do not have a vice, you can continue from here using the same steps, but use something to prop up and immobilize the panel while soldering.

Insert the jacks into the jackboard. Flip the board over so that the solder-side is facing up, reverse the board, and insert them into the panel. Tighten their washers on the panel backwards and upside down as shown below. This will allow you to solder each of the jack boards easily in a way that will also ensure that they will fit properly.

Step 27: (Kit) Solder Jack Boards

Solder the jacks to the board. When you finish soldering the first jack board, remove it, set it aside, and then insert the next set of jacks and repeat the process. Do not solder the jacks to their jackboards without inserting them into the panel as described above — if you solder them outside the panel, they may not fit correctly, and you will have to resolder them to get them to fit. Once you are finished with the second board, remove it and insert both jack boards in their correct orientation. If you are using a vice, do not remove the module just yet.

Step 28: (Kit) Insert and Solder Buss Wires to Jack Boards

Grab 4 of the jumpers you set aside in step 4. Start with the ground buss between the input and output jack (labeled GND). Insert the jumper through both holes and bend the jumper at the top to hold it in place. Solder the jumper to the bottom (Input) jack board first. Now solder the GND pad on the top (Output) jack board. Next, using the same technique, connect the Tip pads (labeled T) between the jack boards for the Bias and Bias Mix jacks. After that, take the 3rd jumper and connect the T pads of the Freq and Freq Mix jacks together. Finally, solder the last jumper between the triangle wave output and the Mod input through the Tip Switch pads (labeled TS). The pictures clearly illustrate each of these connections.

Step 29: Solder the Jack Boards to the PCB

Connect the remaining wires from the jack PCBs to the main PCB. Start with the connections closest to the jack PCBs and work outwards. Below are the specified wire lengths for each connection in the suggested soldering order:

- Square: 2in (5cm) - to Pad V
- Triangle: 2in (5cm) - to Pad U
- Mod: 2in (5cm) - to Pad Q
- Input: 3in (7.5cm) - to Pad E
- Output: 3in (7.5cm) - to Pad A
- Bias: 3in (7.5cm) - to Pad O
- Bias Mix: 3in (7.5cm) - to Pad P
- Freq: 4in (10cm) - to Pad R
- Freq Mix: 4in (10cm) - to Pad S
- Saw: 4in (10cm) - to Pad T

Step 30: Power Up Your Module

Connect your module to your rack's power source and turn it on. Sweep the Freq pot and watch for the blinking LED. If the LED does not turn on, quickly disconnect the module and start troubleshooting (check orientations!). If the LED is flashing, then your module is likely assembled correctly.

Step 31: Test Your Module

To assess your module's functionality, run it through our factory test routine:

1. Feed a noise source or harmonically-rich oscillator waveform (like a Saw) to the module's input.
2. Switch the module to the "Out" (bypassed) position.
3. Feed the output of the module to an amplifier. You should now hear the sound without any modulation. If you hear modulation, check to make sure that your In/Out switch is wired correctly and is also not upside down.

4. Turn the Level knob all the way down. This module can add a lot of gain to your signal, and suddenly switching it in with the Level knob turned all the way up may be an unpleasant surprise.
5. Switch the module into VCA mode, then switch in the effect.
6. Turn the Freq knob down to around 9 o'clock so that it is flashing rapidly, but not so quickly that it looks persistently on.
7. Turn the Depth and Bias knobs all the way up and the Gain knob all the way down.
8. Slowly turn up the Level knob.
9. You should now be hearing a tremolo (amplitude modulating) effect. If you are hearing a filtering effect, make sure the Mode switch is wired correctly and not turned upside down. If you hear nothing, check the wiring on your In/Out switch.
10. Turn up the Gain knob - you should hear the signal begin to distort.
11. Turn down the Bias knob - you should hear the signal become less distorted.
12. Turn the Bias knob back up and then turn the Depth knob down - you should hear the modulation fade out, but the distortion will remain.
13. Switch into VCF mode. The effect should now sound like an auto wah (sweeping bandpass filter). Test all the controls in the same way you did for VCA mode.
14. Turn the Bias all the way down with the Depth all the way up. Turn the Filter Mix knob all the way down. You should now hear the filter sweeping in the high frequency ranges. Slowly turn the Filter Mix knob up - you will begin to hear the low end starting to come back in. If your input signal is very hot and the Gain is turned all the way up, as you turn the Filter Mix knob up, your signal may distort to the point where it sounds like it "ducks out," similar to a sidechained compressor. This is normal and is caused by super-saturating the FET.
15. Turn the Bias knob all the way up, the Filter Mix all the way down, and the Freq knob back into the 9 o'clock range if you have since changed it.
16. Take a patch cord and plug it into the Mod input. You should hear the modulation stop suddenly. This is because you are breaking the normalized connection between the Triangle LFO output jack and the Mod input jack. With the patch cord still connected to the Mod input, insert the other end into the Triangle LFO output. This should sound the same as the module when it is unpatched. If it does not, check the buss wiring between the jack boards.
17. Remove the patch cable from the Triangle output and insert it into the Saw output. You should notice a change both sonically and in the behavior of the LED. If it is modulating but sounds like a square wave, you probably have the wiring reversed.
18. Remove the patch cable from the Saw output and insert it into the Square output. Although the Square output is not a true square wave, it should have a more pronounced on/off modulation effect compared to the Triangle and Saw outputs. If it sounds more like a saw wave, again, you may have reversed the wiring on the jack boards.
19. Switch the module back into VCA mode. Remove the patch cable from the Mod input and place the Square wave LFO into the Freq Mix jack. You should hear some self modulation happening — if you don't, try adjusting the Freq knob to offset the input voltage. If you don't hear anything or the LFO suddenly stops, recheck your jacks' wiring. If the Freq Mix input works, you can be certain that the Freq input works, as they are flowing to the same part of the circuit. You should, however, plug into the Freq input to see that it switches out the Freq knob. With a patch cable plugged into the Freq input, the Freq knob should be disabled.
20. Unplug the Square wave from the Freq Mix and plug it into the Bias Mix input. With the Bias turned all the way up, you should hear some faint modulation in the distortion characteristics. This is normal. Because all of the LFO outputs are synced to one another, the effect of modulating the Bias or Frequency is less pronounced than it would be if modulate those parameters from a different

source. Like the Freq Mix input, you do not need to test the Bias input separately. You should, however, plug into the Bias input to see that it switches out the Bias knob. With a patch cable plugged into the Bias input, the Bias knob should be disabled.

If your module has passed inspection, congratulations! You are finished with construction.

Step 32: Troubleshooting Your Module

Troubleshooting:

If it does not seem to work like it should, these are common reasons why it might not, ranked from most to least likely:

1. Diodes are backwards
2. Electrolytic caps are backwards*
3. LED is backwards
4. Transistors are backwards*
5. ICs are backwards*
6. Wiring is incorrect*
7. Missing the M-N jumper
8. Components were overheated*
9. Power jack is backwards***

* — if you find these problems when troubleshooting your build, you will likely need to desolder and replace some components, because they are probably destroyed. Backwards or incorrect wiring can cause a host of problems — look for nearby damaged caps first then spread out from there.

** — an easy rule to follow with soldering: if it looks terrible, it is terrible. In other words, if you would be angry if you bought this module pre-made and it had the quality of solder work that you just gave it, your problem might be with your soldering. The solution here is to step away from this project and go back to YouTube or down to simpler projects to practice good soldering techniques before returning to repair it. Trying to repair your kit at this point might just do more harm than good. Your solder should flow meaning you shouldn't have to cut off golf balls of solder from the bottom of the PCB. If you are trimming a solder joint that is spherical, it means that the solder has not made a good connection. If you have bad soldering technique, your problems may be myriad and intermittent.

*** — if you soldered the power jack backwards and powered the unit up, you probably did no harm to your Gristleizer — that is, if you correctly oriented the polarity protection diodes. If you didn't install the diodes correctly and fed the module power incorrectly, you probably caused a cascade of failures — at that point it might be worth it to start over with a fresh board. If you did solder the power jack backwards but didn't harm your module, it might be worth your time just to cut out a new keyhole with a razor blade (carefully!) rather than trying to desolder it from the board. You'll have to be mindful in the future about how you connect it though!